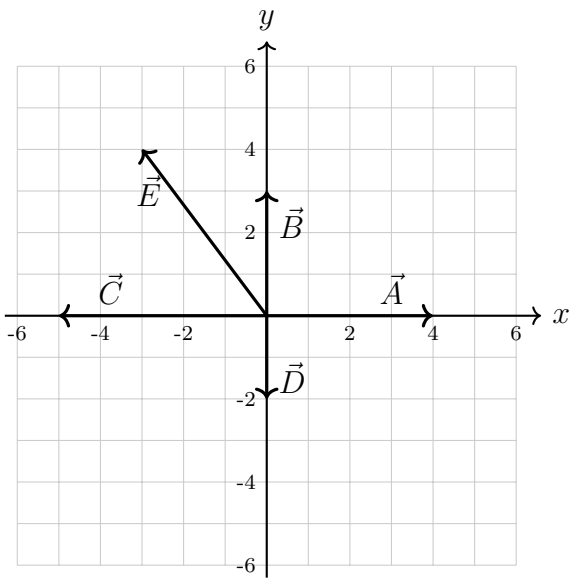
2.1 Homework: Reference Frames and Displacement Solutions

1. Vectors from the origin.

   (a) $\vec{E}$ is **3 m west** and **4 m north**.

   (b) $|\vec{E}| = \sqrt{3^2 + 4^2} = \sqrt{25} = \mathbf{5\ m}$

   Each arrow starts at the origin and the head is at the end. $\vec{E}$ is a 3-4-5 right triangle; accept 5 m measured with a ruler against the grid. $\vec{E}$ is one vector, not two: its two parts only describe it.



2. Vectors drawn anywhere.

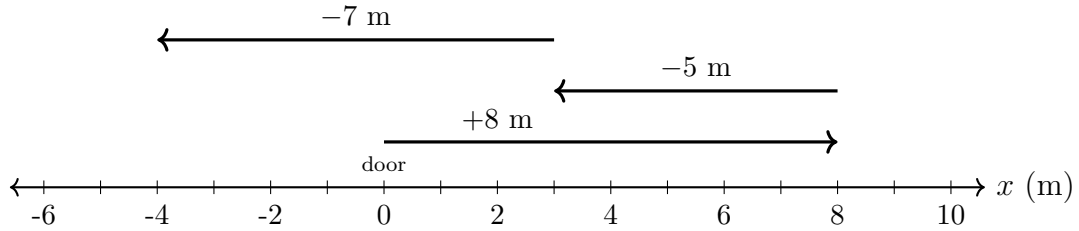
   (a)

	left/right	up/down
$\vec{P}$	**3 right**	**2 up**
$\vec{Q}$	**3 right**	**2 up**
$\vec{R}$	**3 left**	**2 down**
$\vec{S}$	**0**	**3 up**

   (b) $\vec{P}$ **and** $\vec{Q}$. *Same length and same direction. They start at different points, but where a vector is drawn does not matter.*

   (c) $\vec{R}$. *Same length, opposite direction: $\vec{R} = -\vec{P}$. A student who answers $\vec{S}$ is judging by eye; ask them to count squares.*

3. Back and forth along the hallway.



   (a) $0 + 8 - 5 - 7 = -4$, so $x = \mathbf{-4\ m}$ *(4 m west of the door)*

   (b) Displacement $= -4\text{ m} - 0\text{ m} = \mathbf{-4\ m}$, **that is, 4 m west**

   (c) Distance $= 8 + 5 + 7 = \mathbf{20\ m}$

Accept "4 m west" or "−4 m" for (b), but not "4 m" alone: a displacement needs its direction. Distance adds the lengths and ignores direction, so it is never negative.

4. Distance and displacement.

Trip	**Distance**	**Displacement**
(a) One lap of a 400 m running track	**400 m**	**0**
(b) Empire State Building: 320 m up to floor 86	**320 m**	**320 m up**
(c) Up to floor 86, then back down to the lobby	**640 m**	**0**
(d) Manhattan: 10 blocks north, then 4 blocks south	$\approx$ **1120 m**	$\approx$ **480 m north**

(d): distance $= 10 + 4 = 14$ *blocks* $\approx 14 \times 80 = 1120$ *m; displacement* $= 10 - 4 = 6$ *blocks north* $\approx 6 \times 80 = 480$ *m north. Accept answers left in blocks, and accept "about 1100 m"; the block length is only approximate. In (a) and (c) the trip ends where it started, so the displacement is zero however far the person went.*

5. Reference frames.

   (a) **0 km/h** relative to the train; **300 km/h** relative to the ground.

   Both are correct. A speed means nothing until we say what it is measured against.

   (b) Toward the front: $300 + 5 =$ **305 km/h north**

   Toward the back: $300 - 5 =$ **295 km/h north**

   Take north as positive: the walk is $+5$ *km/h or* -5 *km/h relative to the train, added to the train's* $+300$ *km/h. Walking toward the back, the passenger still moves north over the ground — just a little more slowly than the train.*

   (c) $300 + 300 =$ **600 km/h (going south)**

   In the seated passenger's frame the passenger is at rest and the ground itself moves south at 300 km/h; the other train moves south at 300 km/h relative to that ground, so 600 km/h in all. This is why two trains passing feels so violent, even though neither is going faster than 300 km/h.